

ML Problem Framing Framework

Overview

Most ML projects follow this sequence:

Business Goal → **Decision to Improve** → **Prediction/Optimization Task** → **Data**
→ **Model** → **Deployment** → **Business KPI**
(KPI = Key Performance Indicator)

Example

- **Business Goal:** Reduce customer churn
- **Decision:** Which customers should receive retention offers?
- **ML Task:** Predict probability of churn
- **Data:** Usage history, complaints, payments
- **Model:** Classification model
- **Deployment:** Daily scoring pipeline
- **KPI:** Reduced churn rate, increased revenue

1. Customer Churn Prediction

Business Problem

Telecom, SaaS (Software as a Service), Banking, and Insurance companies lose customers every month. Losing an existing customer is often much more expensive than retaining one.

ML Problem Framing

Predict whether a customer will leave within the next X days.

ML Type

Classification.

$$Y = \begin{cases} 1 & \text{Customer churns} \\ 0 & \text{Customer stays} \end{cases}$$

Features

- Usage frequency
- Login behavior
- Payment history
- Complaints
- Support tickets
- Subscription age

Industry Implementation

Companies like Netflix, Spotify, telecom operators, and SaaS firms run churn models daily.

Output — Customer Churn Score

Customer	Churn Probability
A	92%
B	18%

Marketing team targets Customer A.

2. Fraud Detection

Business Problem

Banks and payment platforms lose money due to fraudulent transactions.

ML Problem Framing

Predict whether a transaction is fraudulent.

ML Type

Binary Classification + Anomaly Detection.

Target: Fraud / Non-Fraud.

Features

- Transaction amount
- Location
- Device ID
- Time of day
- Merchant category
- Historical behavior

Industry Implementation

Used by Visa, Mastercard, banks, and fintechs.

Real Pipeline

```
Transaction arrives
↓
Feature Store
↓
Model Inference (<100ms)
↓
Approve / Reject / Manual Review
```

3. Credit Risk Modeling

Business Problem

Should we give a loan?

ML Problem Framing

Predict probability of default.

ML Type

Classification.

Target: Default / No Default.

Features

- Income
- Employment history
- Credit score
- Existing loans
- Banking behavior

Industry KPI

- Lower defaults
- Higher loan approval quality

Common Models

- Logistic Regression
- XGBoost
- LightGBM

4. Demand Forecasting

Business Problem

How much inventory should be stocked?

ML Problem Framing

Predict future demand.

ML Type

Time Series Forecasting.

Target: Future sales quantity.

Features

- Historical sales
- Holidays
- Promotions
- Weather
- Seasonality

Industry Examples

Retail: Walmart, Amazon.

Manufacturing: Production planning, inventory optimization.

KPI

Reduce stockouts and overstocking.

5. Recommendation Systems

Business Problem

Which product/content should be shown next?

ML Problem Framing

Predict user–item preference score.

ML Type

Ranking / Recommendation.

Data

User: Clicks, purchases, watch history.

Item: Category, price, metadata.

Industry Examples

YouTube, Amazon, Netflix.

Pipeline

```
Candidate Generation
↓
Ranking Model
↓
Top-K Recommendation
```

6. Predictive Maintenance

Business Problem

When will a machine fail?

ML Problem Framing

Predict failure before breakdown.

ML Type

Classification + Time Series.

Data — IoT Sensors

- Temperature
- Vibration
- Pressure
- RPM

Industry

Manufacturing, Aviation, Energy.

KPI

Reduce downtime.

Example

Aircraft engine monitoring by companies such as GE Aerospace.

7. Dynamic Pricing

Business Problem

What should today's price be?

ML Problem Framing

Predict optimal price maximizing revenue/profit.

ML Type

Regression + Optimization.

Features

- Demand
- Competition
- Inventory
- Seasonality

Industry Examples

Uber surge pricing, airlines, hotels.

KPI

Revenue uplift.

8. Customer Lifetime Value (CLV)

Business Problem

How valuable will this customer become?

ML Problem Framing

Predict future revenue generated by a customer.

ML Type

Regression.

Target: Future 12-month revenue.

Applications

- Marketing budget allocation
- Acquisition targeting

9. Lead Scoring

Business Problem

Which leads should sales call first?

ML Problem Framing

Predict probability of conversion.

ML Type

Classification.

Features

- Company size
- Industry
- Website visits
- Email engagement

Industry

B2B SaaS companies.

KPI

Increase sales productivity.

10. Computer Vision Quality Inspection

Business Problem

Detect defective products automatically.

ML Problem Framing

Image Classification / Object Detection.

Examples — Manufacturing

- Scratches
- Cracks
- Missing components

Pipeline

```
Camera
↓
Image Processing
↓
Defect Detection Model
↓
Accept / Reject
```

11. NLP Customer Support Automation

Business Problem

Reduce support workload.

ML Problem Framing

Classify or answer customer queries.

ML Tasks

- Intent Classification
- Ticket Routing
- Chatbots
- RAG Systems

Industry Examples

Banks, telecoms, e-commerce companies.

KPI

- Faster resolution
- Lower support cost

12. Supply Chain Optimization

Business Problem

How should goods move through the network?

ML Problem Framing

Predict + Optimize.

Components

```
Forecast demand
↓
Predict delays
↓
Optimize routing
```

Industry

Logistics companies such as DHL and FedEx.

How ML Projects Actually Look in Industry

A common misconception is:

Business Problem → Deep Learning

Reality is:

Stage	Time Share
Problem Framing	15%
Data Collection	25%
Data Cleaning	30%
Feature Engineering	15%
Modeling	10%
Deployment & Monitoring	5%

Many successful production systems still use:

- Logistic Regression
- Random Forest
- XGBoost
- LightGBM

rather than complex deep learning models.

Interview-Oriented Problem Framing Template

When given any business case:

1. Define business objective.
2. Define decision to be improved.
3. Define ML task:
 - Classification?
 - Regression?
 - Forecasting?
 - Ranking?
 - Clustering?
4. Define target variable.
5. Define available features.
6. Define evaluation metric.
7. Define deployment strategy.
8. Define business KPI.

Example

“An e-commerce company wants to reduce returns.”

- **Objective:** Reduce return losses.
- **Decision:** Which orders are likely to be returned?
- **ML Task:** Classification.
- **Target:** Returned / Not Returned.
- **Features:** Product type, customer history, price, delivery time.
- **Metric:** Precision, Recall, F1.
- **Deployment:** Score orders before shipment.
- **KPI:** Reduced return cost.

This is the mindset used by ML engineers, data scientists, and applied AI teams when converting real business problems into deployable machine-learning solutions.